Fiscal Year:	FY 2007	**Task Last Updated:**	FY 08/01/2007
PI Name:	Nelson, Gregory A. Ph.D.		
Project Title:	NSCOR: Progressive Alterations of Central Nervous System Structure and Function Are Caused by Charged Particle Radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92350-1700	**Congressional District:**	31
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSCOR 03-OBPR-02
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	End date changed to 2/28/2009 per PI (5/08) End date changed to 1/23/2009 per JSC info update (7/07)		
Key Personnel Changes/Previous PI:	Weili Lin, Ph.D. University of North Carolina. Subcontract terminated January 23, 2007. ; Marco Stampanoni, Ph.D. Paul Scherrer Institute, Villigen, Switzerland. New subcontract ; Lead Investigator for X-ray Tomaography. New Subcontract November, 2006.		
COI Name (Institution):	Archambeau, John O. (Loma Linda University) Chang, Polly Y. (SRI International) Fike, John R. (University of California San Francisco) Krucker, Thomas (Novartis Institute for Biomedical Research) Lin, Weili (University of North Carolina at Chapel Hill) Mao, Xao-Wen (Loma Linda University) Obenaus, Andre (Loma Linda University) Pecaut, Michael J (Loma Linda University) Song, Sheng-Kwei (Washington University School of Medicine) Spigelman, Igor (University of California at Los Angeles) Stampanoni, Marco Ph.D. (Paul Scherrer Institute)		

Grant/Contract No.:	NNJ04HC90G
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Task Description:	The experimental program detailed above will provide a comprehensive, quantitative assessment of the effects of charged particle radiation on the mammalian brain using the mouse hippocampus as a model. The mouse brain will be irradiated with accelerated protons and iron ions and examined over the course of two years for progressive changes in hippocampal structure and function. Cell populations in tissue sections will be quantified by a high precision stereology method to measure rates of loss of neurons, vascular endothelial cells and glia. Brain blood vessel topology will be further characterized by an erosion casting technique that produces a 3-D map of the vessels at 1 micron resolution. Stem cells and ongoing neurogenesis will be quantified using a complementary technique in which dividing cells are directly detected. Activation of cells and residual genetic damage in these cells will be characterized by immunofluorescence and mRNA level measurements of selected cytokine biomarkers and analysis of mutations in the transgene b-galactosidase. Functional impairments of neuronal ensembles will be quantified by electrophysiological measurements of hippocampal tissue slices and patch clamp measurements will detail changes in membrane properties. System level changes will be quantified by electroencephalography and two magnetic resonance imaging modalities that map white matter tracts (diffusion tensor imaging) and local metabolic rate (oxygen extraction). The plasticity of the brain as a system will be tested in its reaction to an immune system challenge mediated by lipopolysacharide. Finally the ability of radiation to modulate the latency of a neurodegeneration process will be tested in a transgenic mouse that overproduced amyloid precursor protein and develops an Alzheimer disease-like pathology. Taken together these results should provide a basis for determining risk to the CNS from charged particle radiation exposure. The main hypothesis for the grant is: Charged particle radiation exposure induces progressive deleterious structural and functional changes in the brain, leading to performance decline. Research Aims: Hypothesis (A) Charged particle irradiation causes a progressive loss of cells and a remodeling of CNS tissue as a function of dose, time and LET. To test hypothesis A we will: Specific Aim A1. Quantify the cellular composition of mouse brain vascular endothelium using stereological analysis of preserved tissue. Specific Aim A2. Quantify the composition of mouse neurons, selected glia and stem cells using stereological analysis of preserved tissue. Specific Aim A3. Quantify the activation of resident and infiltrating mouse immune cells using stereological analysis immune-labeled tissue. Specific Aim A4. Quantify key molecular biomarkers associated with radiation-induced damage to mouse cells and their microenvironment using targeted gene expression profiling and mutation analysis of a structural gene. Hypothesis (B) Charged particle irradiation alters the functional output or performance of the brain as a function of dose, time and LET. To test hypothesis B we will: Specific Aim B1. Quantify tissue-level electrophysiological properties of mouse brain using tissue slices that represent intact regional ensembles of interacting cells. (extracellular, patch clamp) Specific Aim B2. Map the time evolution of macroscopic structural features and metabolic alteration in mouse brains using MRI. Hypothesis (C) Exposure to radiation decreases disease latency and alters response to acute stressors. To test hypothesis C we will: Specific Aim C1. Quantify the magnitude and time course of electrophysiological changes in brain slices of mice expressing a human transgene that produces an Alzheimer disease-like pathology and limits output. Specific Aim C2. Quantify the electrophysiological performance of brain slices in mice reacting to the application of systemic immune stress using lipopolysaccharide as a surrogate infectious agent Foremost in the Critical Path Roadmap system for radiation effects on the central nervous system is Risk #29 – Acute and Late CNS Risks. All of the research and technology questions associated with this risk have priority level one for Mars missions and priority level 3 levels for ISS and 30d lunar missions. The questions addressed by the NSCOR project are 29a (functional changes), 29b (late degeneration), 29c (Alzheimer predisposition), 29e (late degeneration), 29g (cell and tissue mechanisms of damage), 29h (different cell populations including stem cells) and 29i (biomarkers). Also, a NASA workshop on CNS radiobiology identified supporting questions to assist in designing experiments to elucidate CNS effects in response to radiation exposure. These questions include: 1) What is the latency to CNS injury, 2) What cell types are involved in CNS injury (neurons, stem cells, glia)? and 3) What vascular changes occur in response to exposure. We have used this framework to focus our investigation on the effects of charged particles on the CNS. Specifically we will quantify cellular and functional changes in the hippocampus following high energy protons (main SPE constituent) and Fe-56 particles . We will use male rodents with sufficient statistical samples to address inter-individual variation and will address workshop questions 1-3 by examining the responses of different constituent cell types and the latency of damage. NSCOR subproject A quantifies different cell populations (29b, 29e, 29g, 29h); subproject B addresses functional changes and non-invasive techniques for identifying biomarkers (29a, 29i); and subproject C targets late degenerative changes with an Alzheimer model and the role of inflammation (29b, 29c, 29e, 29g and 29i). Our strategy is therefore aligned with the programmatic goals set forth by NASA and its advisory groups.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The response of the central nervous system to ionizing radiation will have direct impact on the development of radiotherapy using charged particles. The ability of protons, helium ions and carbon ions to be localized precisely in treatment fields close to critical structures (eg. optic nerve) make charged particle radiotherapy highly desireable. The limiting biological constraint on such treatments is the reaction of normal tissue surrounding a tumor site. The results of the NSCOR task will quantify cell loss and tissue functional changes form proton and iron irradiation that can be used directly to estimate normal tissue complication probabilities in the context of radiotherapy.
	Administrative Changes. Two significant administrative changes occurred during project year 3. First, an Aim B subtask has been terminated along with the associated sub-contract to develop MRI sequences enabling the mapping of brain metabolic levels indicated by the relative levels of oxy- vs. deoxyhemoglobin. The sequences were established on a clinical system, then successfully rewritten for a Brucker clinical system and transferred to Loma Linda. However, the contractor was unable, in a timely fashion, to configure the software for our Brucker 4.7T and 11.7 T micro-imagers needed for the project. As irradiated animals would no longer be available for imaging before the final software was completed it was decided to terminate the task. The second significant change was the completion of sub-contractual arrangements for the Paul Scherrer Institute in Villigen, Switzerland to provide synchrotron micro-tomography services for 3-D reconstruction of brain vascular corrosion casts. The task officially began in November 2006 under the direction of Dr. Marco Stampanoni.

Task Progress:

Technical Progress

Irradiations: Most of the animals scheduled for iron irradiation at the Brookhaven National Laboratory have now been completed and are aging to their appropriate harvest dates. An additional group of transgenic animals are scheduled for irradiation at NSRL on October 17, 2007. All Aim B animals have been irradiated with both iron ions and protons. Proton irradiations of animals for stereological analysis under Aim A were completed in three campaigns: 12/13/06, 1/9/07 and 1/13/07. A final group of animals for mutagenesis experiments (Aim A4) will be irradiated in late summer / early fall 2007 as determined by the breeding schedule of transgenic LacZ mice.

Core Activities: Tissue processing, stereological and immunohistochemical analyses continue under the supervision of Dr. C. Favre. Methods have continuously improved and staff have been trained to use a new Olympus FV1000 Confocal Microscope system that complements an existing Biorad system. This has enhanced the throughput of image acquisition from prepared samples. Several neurogenesis-related samples (Aim A2) have been repeated to eliminate interference on image quality due to tomato lectin labeling for lumens of microvessels. Vascular corrosion casting training was completed and an LLU team is now producing vascular corrosion casts which are being imaged by X-ray micro-tomography at the Paul Scherrer Institute. The central database is operating stably using the Linux operating system in its permanent location.

Aim A Activities.

Microvasculature Stereology. Stereological analysis of iron-irradiated mice is continuing on the set of 230 brains which have been preserved and archived. The systematic sections cut from each brain slice containing compartments of interest were stained to identify the capillaries, and cell nuclei of interest. Microvessel endothelial cell population numbers as well as vessel lengths in CA1 and dentate gyrus (DG) regions of the hippocampus for times up to 12 months following 56Fe have been measured with no significant reduction of endothelial cell density. However a U-shaped dose response curve is emerging for vessel density at 12 months post irradiation possibly reflecting a repopulation by endothelial cells after higher doses.

Neurogenesis. New sets of animals for neurogenesis have been irradiated with iron ions and analyzed by J. Fike et al. at U.C. San Francisco. This set of animals was a repeat of previous exposures but the signal to noise ratio for neuroblast quantitation was improved by elimination of dual labeling with lectin (used to stain microvessel lumens). These counts have been completed and dose dependent reductions in neuron, astrocyte and oligodendrocyte lineages have been observed. In contrast, an increase in newly-born activated microglia was observed.

Biomarkers of Genotoxicity. For assessment of DNA damage and repair, hippocampal tissues were isolated from animals exposed to a range of iron doses at 1, 8 and 16 weeks after irradiation and processed for lacZ transgene mutant frequencies (MF). We observed a trend of increasing MF with increasing radiation dose up to 4 Gy. Analyses of characteristics of mutants obtained from the 8 week samples using restriction fragment length polymorphism (RFLP) shows that the proportions of mutants with size changes and insertions (versus point mutations) changes substantially with dose and suggests that only cells containing small changes in the genome continue to survive after higher doses. Targeted gene expression studies have highlighted time and dose-related changes in the neural cell adhesion molecule (NCAM) and brain derived neurotrophic factor (BDNF) which are involved in synaptogenesis and neurogenesis.

Aim B activities.

Electrophysiology: Extracellular recordings on 400 um thick hippocampal sections have now been obtained for times up to 12 mo. after iron irradiation. Extracellular recordings were performed in the CA1 dendritic region and after input-output curve characterization, long term potentiation (LTP) was induced by tetanic stimulation with recordings for 90 min post LTP induction. Hippocampal LTP was readily induced in 0 Gy animals but this contrasts with head-only irradiated animals at 1 –12 mo. These animals exhibited dose-dependent decrements in post-tetanic potentiation (PTP) and LTP was decreased significantly at 2 and 4 Gy. These results demonstrate that hippocampal plasticity is altered after 56Fe head-only irradiation. Patch clamp analysis of individual nerves has been carried out at UCLA for animals at 3, 6 and 12 months post irradiation but no obvious changes in intrinsic membrane properties were detected. Neuropharmacological tests are now planned to investigate aspects of neurochemical transmission and receptor function.

Magnetic Resonance Imaging. Diffusion tensor imaging (DTI) magnetization sequences and data analysis software have now been transferred to LLU from Washington University and animals are now imaged on-site. Programs for DTI map calculation were applied to water phantom images to optimize signal to noise ratios and to validate custom coil performance. In vivo and ex vivo images have now been acquired for irradiated animals and show changes in fiber tract morphology and function as reported by the relative anisotropy index. Animal cohorts are being aged appropriately for later image acquisition.

BOLD imaging sequences that portray cerebral metabolic rate of oxygen utilization were adapted from a clinical Siemens system to a clinical Brucker MRI system in use at LLU but the software components were not adapted to the systems required for microimaging of mice. The delays in BOLD imaging implementation precluded their use on aging, irradiated animals so the subcontract was not renewed and this task was terminated.

Aim C Activities.

Electrophysiology. In immunologically-challenged mice (lipopolysaccharide teatment) the most significant finding was the inhibitory effect of radiation on excitability of CA1 neurons. This was detected as a decrease of the amplitude of population spikes evoked at maximal stimulation intensity. Paradoxically, in mice previously exposed to iron ions at 0 to 4 Gy, the effects of irradiation abrogated the LPS-induced response suggesting that irradiation alters the ability of hippocampal structures to respond appropriately to the immune stresses. These observations are consistent for times up to 12 months post-irradiation and are maximal at 2 Gy Corrosion Casting Vascular corrosion casting of transgenic mice began in September 2006 following on-site training at LLU. Corrosion casts of APP23 mouse brains are now routinely made and processed with an X-ray contrast agent. Casts have been shipped to the Paul Scherrer Institute whole brain and local 3-D computed tomography maps of the brain vasculature have been made for younger animal cohorts. Regions of interest have been selected for high resolution mapping and data sets for biological analysis will soon be available for analysis at LLU.

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